



Quantifying Action Uncertainty with Inaccurate Stochastic Dynamics through Conformalized Nonholonomic Lie groups

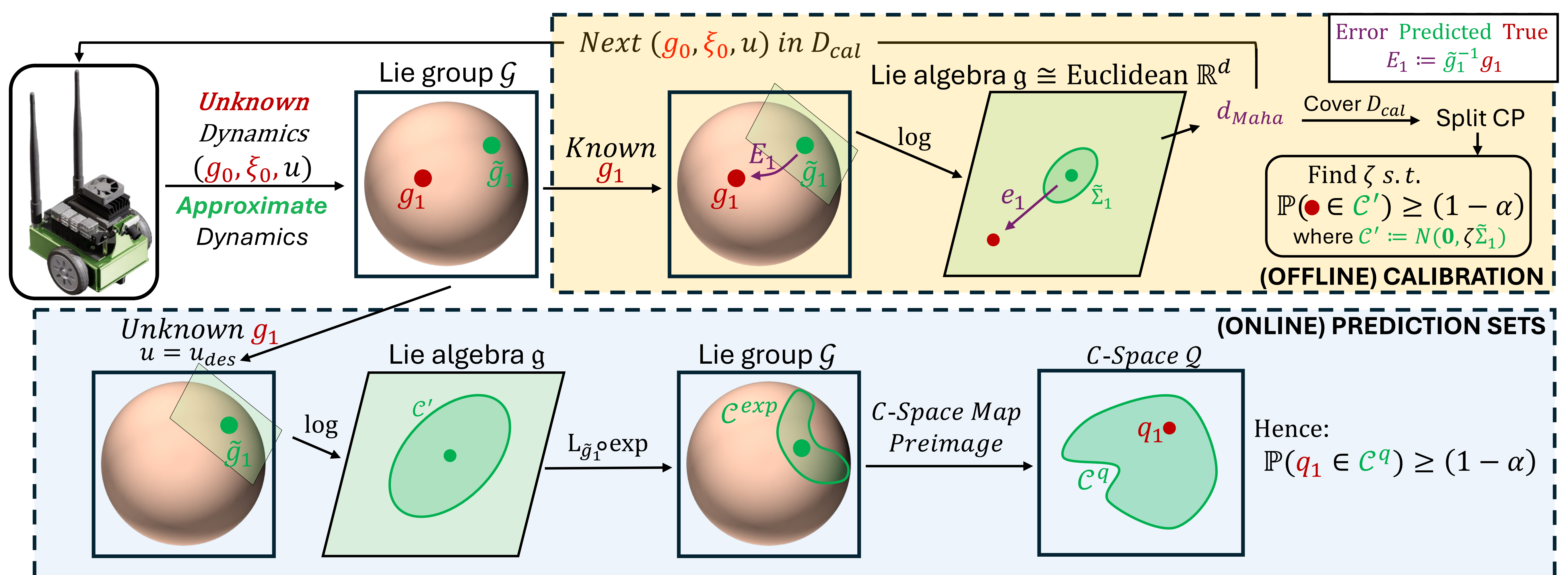
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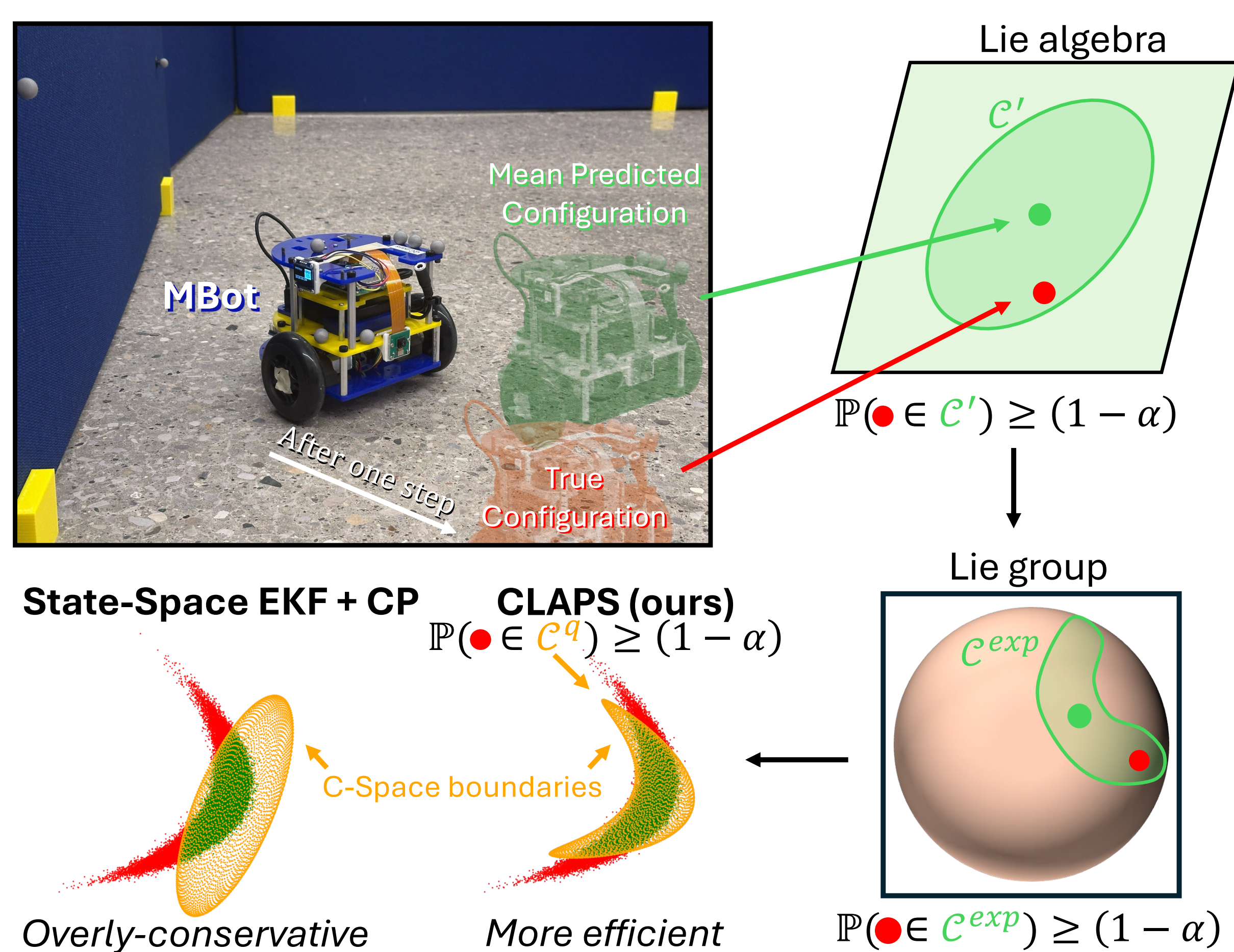
Given an approximate Lie-algebraic Gaussian dynamics model $\tilde{f}$ and a transitions dataset $D_{cal} = \{(s_t, a_t, s_{t+1})\}$, we:

- ▶ provide a state-action-dependent **post-hoc calibration layer** over existing Lie-algebraic Gaussian uncertainty estimators
- ▶ **extend conformal guarantees** from Euclidean configuration spaces to $SE(2)$, improving **volume-efficiency**

Method: CLAPS



Hardware Validation



State Space Dynamics to Lie group form

State Space Dynamics

Configuration Update $\frac{d}{dt} q = \dot{q}$

Forced Lagrange-d'Alembert $\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} = \frac{\partial L}{\partial q} + B(q)u + A(q)\lambda, A(q)\dot{q} = 0$

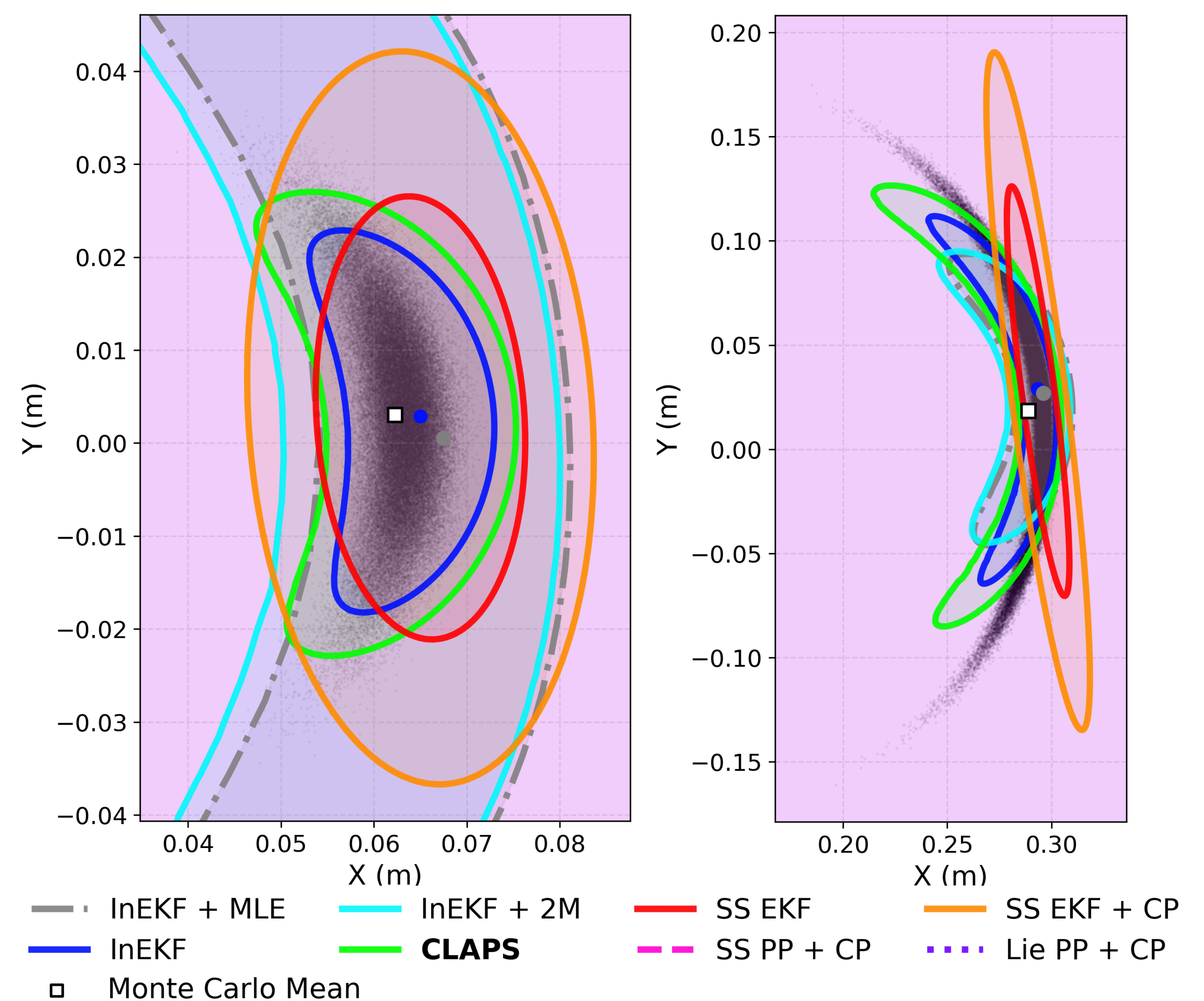
Configuration q / g Velocity $\dot{q} / \xi$ Force Map $B(q) / \mathcal{B}$ Control input u (shared) Nonholonomic Constraint $A(q) / \mathcal{A}$

Lie group Dynamics

Reconstruction Eq. $\frac{d}{dt} g = g\xi^\wedge$

Forced Euler-Poincaré-Suslov $\frac{d}{dt} \frac{\partial l}{\partial \xi} = \text{ad}_\xi^* \left(\frac{\partial l}{\partial \xi} \right) + \mathcal{B}u + \mathcal{A}\lambda, \mathcal{A}\xi = 0$

Workspace Footprint of C-space regions



JetBot (Simulation) Results (over 625 validation trials)

Algorithm	Marginal Coverage (%)	Avg. Volume Ratio ↓	Avg. Workspace IoU with Particles (%) ↑	Provable Guarantees?
SS EKF	78.7	0.63	35.0	✗
InEKF	82.7	0.47	41.8	✗
InEKF+2M	89.2	3.06	40.0	✗
InEKF+MLE	90.3	2.80	42.3	✗
SS PP + CP	89.9	2137	0.20	✓
Lie PP + CP	89.9	2138	0.20	✓
SS EKF + CP	91.2	2.86	30.4	✓
CLAPS	90.0	1.00	48.4	✓

red if coverage does not achieve ($<$) the user-set probability $(1 - \alpha) = 0.9$. The average volume ratio is reported relative to CLAPS.